

# **PROJECT 06 REPORT**

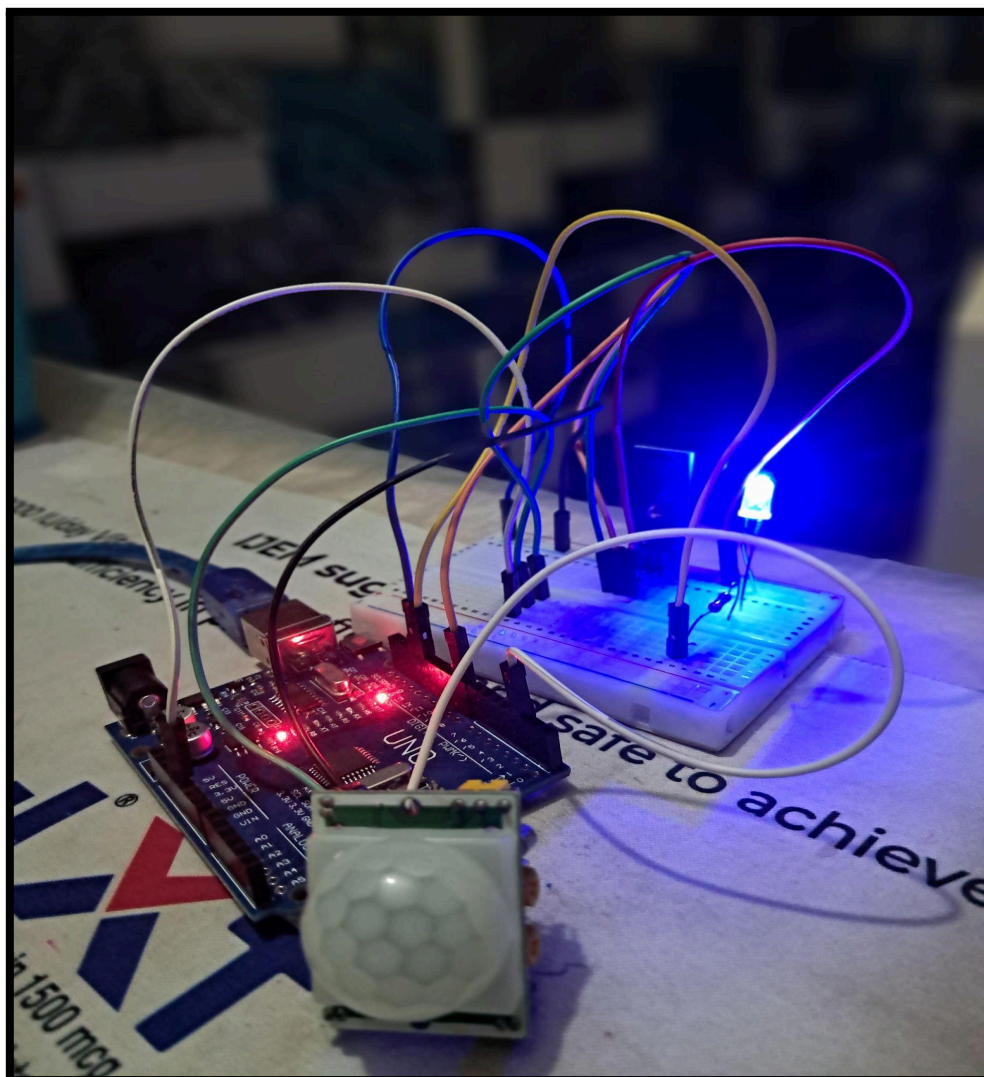
## **Motion Detection Security System using PIR Sensor**

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**Duration :** 2 days



# 1. Objective

The objective of this project was to develop a motion detection security system using a PIR (Passive Infrared) sensor and Arduino Uno.

The project also introduced passive infrared sensing, motion detection, retriggering modes, sensitivity adjustment, delay control and security system design.

# 2. Components Used

| Component            | Quantity |
|----------------------|----------|
| Arduino Uno R3       | 1        |
| Breadboard           | 1        |
| PIR Motion Sensor    | 1        |
| Active Buzzer Module | 1        |
| LEDs (Blue)          | 1        |
| Resistors(300Ω)      | 1        |
| Jumper Wires         | Multiple |
| USB Cable            | 1        |

### 3. Software Used

- Arduino IDE
- Serial Monitor
- Windows Laptop

### 4. Theory

A PIR (Passive Infrared) sensor detects motion by sensing changes in the infrared radiation emitted by warm objects such as humans and animals.

A PIR sensor does not emit any signal but passively monitors infrared radiation in its field of view.

The sensor consists of a PIR sensing element, a Fresnel lens, a BISS0001 controller IC, sensitivity and delay adjustment potentiometers and an H/L mode selection jumper.

The PIR module provides a digital output:

- **No Motion → LOW (0)**
- **Motion Detected → HIGH (1)**

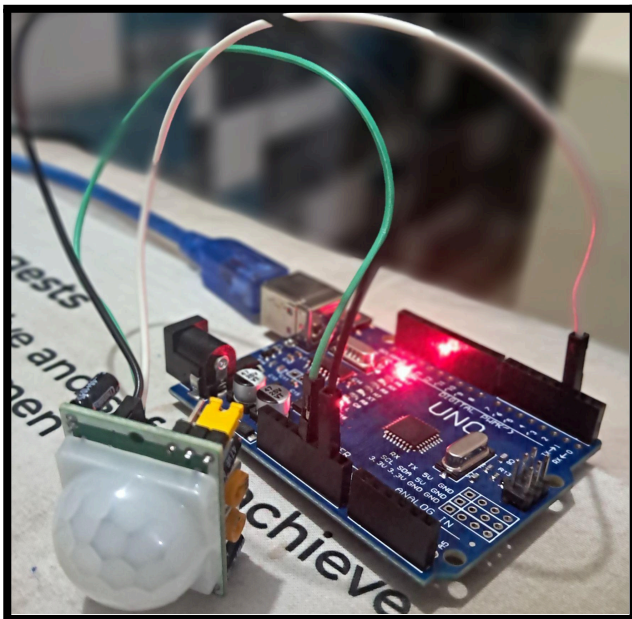
The output can be read using: **digitalRead(2);**

## 5. Tasks Performed

### ➤ Understanding the PIR Sensor

The PIR sensor module was studied to understand its working principle, pin configuration. The concept of passive infrared sensing and motion detection was explored.

### ➤ Reading PIR Sensor Output



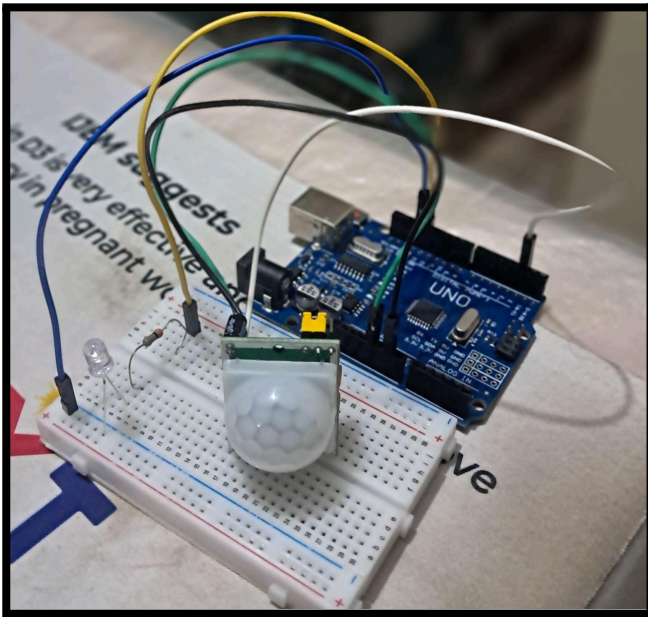
**Video**

The digital output of the PIR sensor was connected to Arduino and monitored through the Serial Monitor using **digitalRead()**.

Initially, the sensor operated in L (Non Retriggering) mode, where the output remained HIGH only briefly after motion detection.

The H/L jumper was then switched to H (Retriggering) mode, allowing the output to remain HIGH while continuous motion was detected.

## ➤ Motion Detection using LED

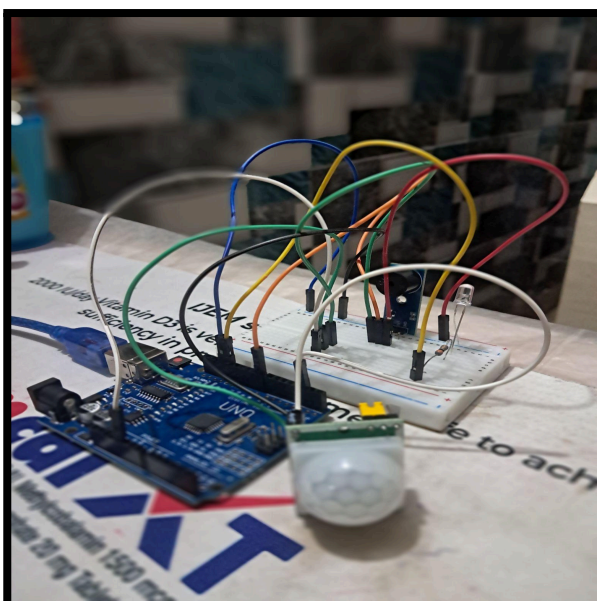


Video

A simple motion detection system was implemented using the PIR sensor and an LED.

Whenever motion was detected, the LED turned ON. Once the motion stopped and the delay timer expired, the LED automatically turned OFF.

## ➤ Motion Detection Security System



Video

The motion detection system was upgraded by integrating both an LED and an active buzzer.

System operation:

- Motion Detected
  - LED ON
  - Buzzer ON
- No Motion
  - LED OFF
  - Buzzer OFF

## 6. Learning Outcomes

In making of this project, the following concepts were learned:

- PIR Sensor Operation
- Passive Infrared Detection
- Motion Detection Principles
- Digital Sensor Interfacing
- H/L Trigger Modes
- Delay Time Adjustment
- Sensitivity Adjustment
- Digital Input Processing
- Motion Detection Alarm Systems
- Embedded Hardware Debugging
- Security System Design



## 7. Problems Faced

- I. Initially, the PIR sensor output remained HIGH only briefly during motion detection. **(The sensor was configured in L mode and the H/L jumper was moved to H mode)**
- II. The output remained HIGH for several seconds even after motion stopped. **(The delay potentiometer was adjusted to reduce the output duration accordingly)**

## 8. Observations

- The PIR sensor detects changes in infrared radiation rather than the presence of a stationary object.
- Continuous motion keeps the output HIGH in H mode.
- The delay potentiometer controls how long the output remains HIGH after motion stops.
- The sensitivity potentiometer adjusts the motion detection range.
- The Fresnel lens(Dome Shape) increases the detection area by focusing infrared radiation onto the sensing element.
- Motion detection was successfully demonstrated using both hand movement and a moving object.
- The security system activated both visual and audible alerts whenever motion was detected.

## **9. Applications and Future Scope**

PIR sensors are widely used in home security alarms, smart buildings, motion activated devices, occupancy detection systems and energy saving automation.

This project can be further enhanced by integrating wireless communication using ESP32, adding camera modules for image capture, implementing mobile notifications and developing complete IoT based home security systems.

## **10. Conclusion**

This project successfully demonstrated the development of a motion detection security system using a PIR sensor and Arduino Uno.

The successful implementation of LED and buzzer alerts demonstrated how PIR sensors can be used in real world security and automation applications, providing a strong foundation for future smart home and IoT based monitoring systems.